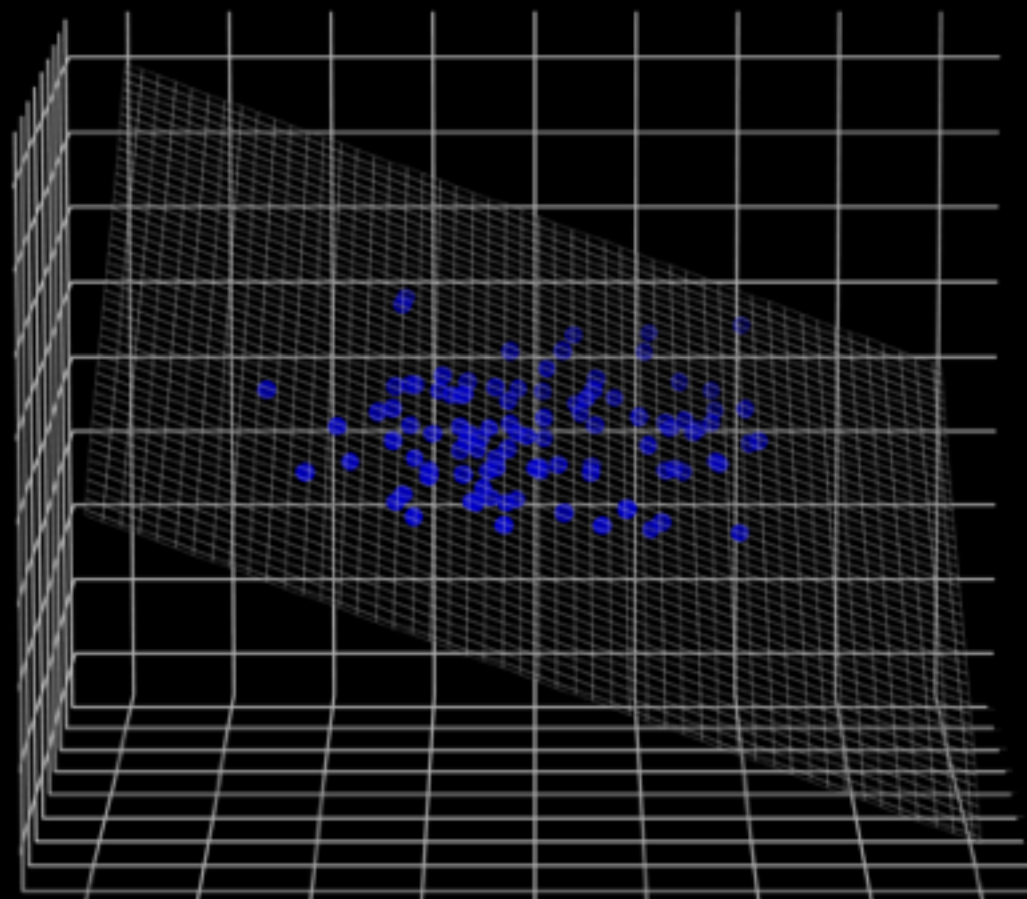
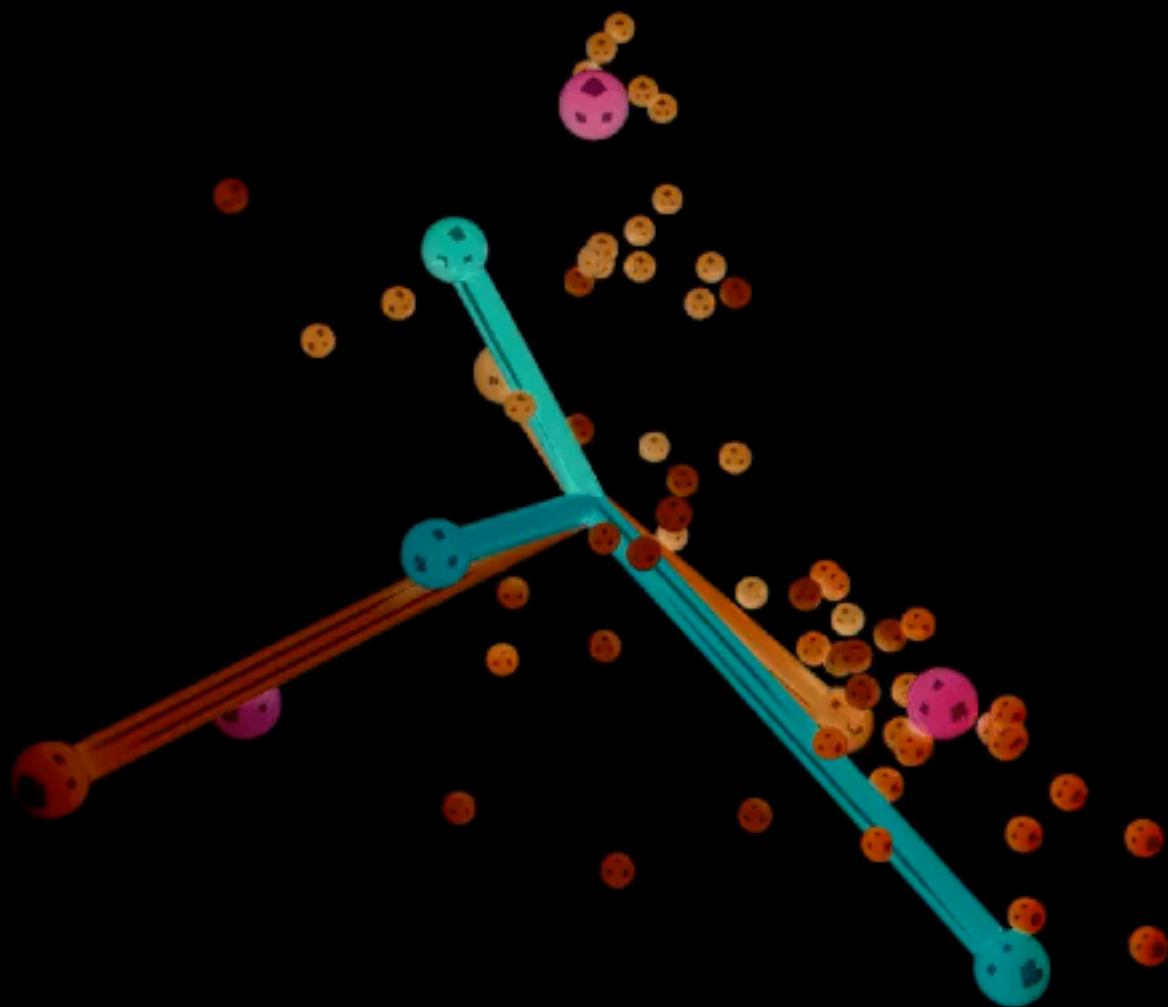








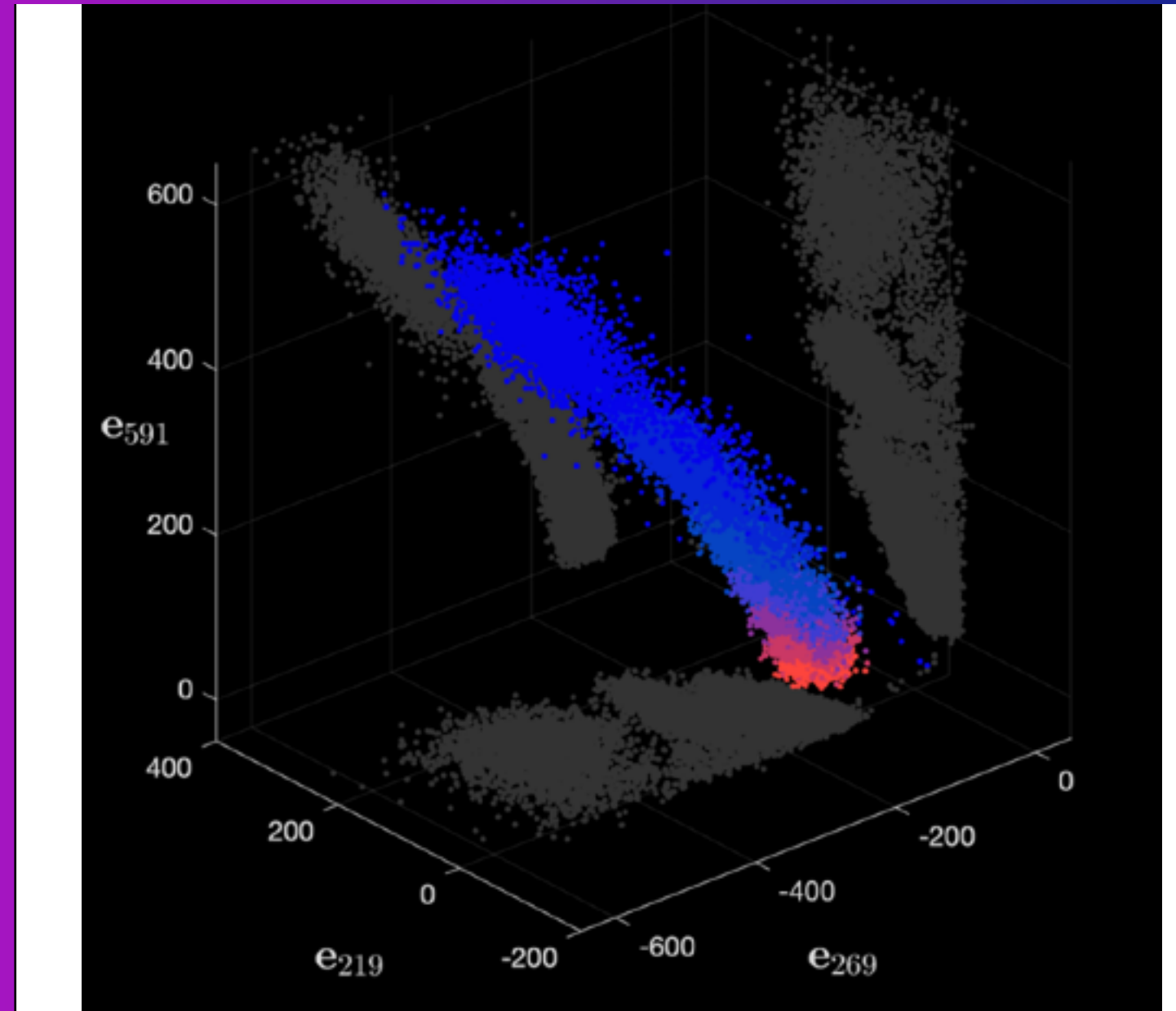
Neural Collapse



Terminal Phase Separation

- ↪ VGG, ResNet, DenseNet, Transformers
- ↪ Theoretical justification through Average Gradient Outer Product
 - ↪ Beaglehole et. al.

Lines of Thought



Stochastic Trajectory prediction

- ↪ Cornell University; Sarfati et. al.
- ↪ Langevin dynamics for continuous dynamical trajectories
- ↪ Fokker-Plank formulation

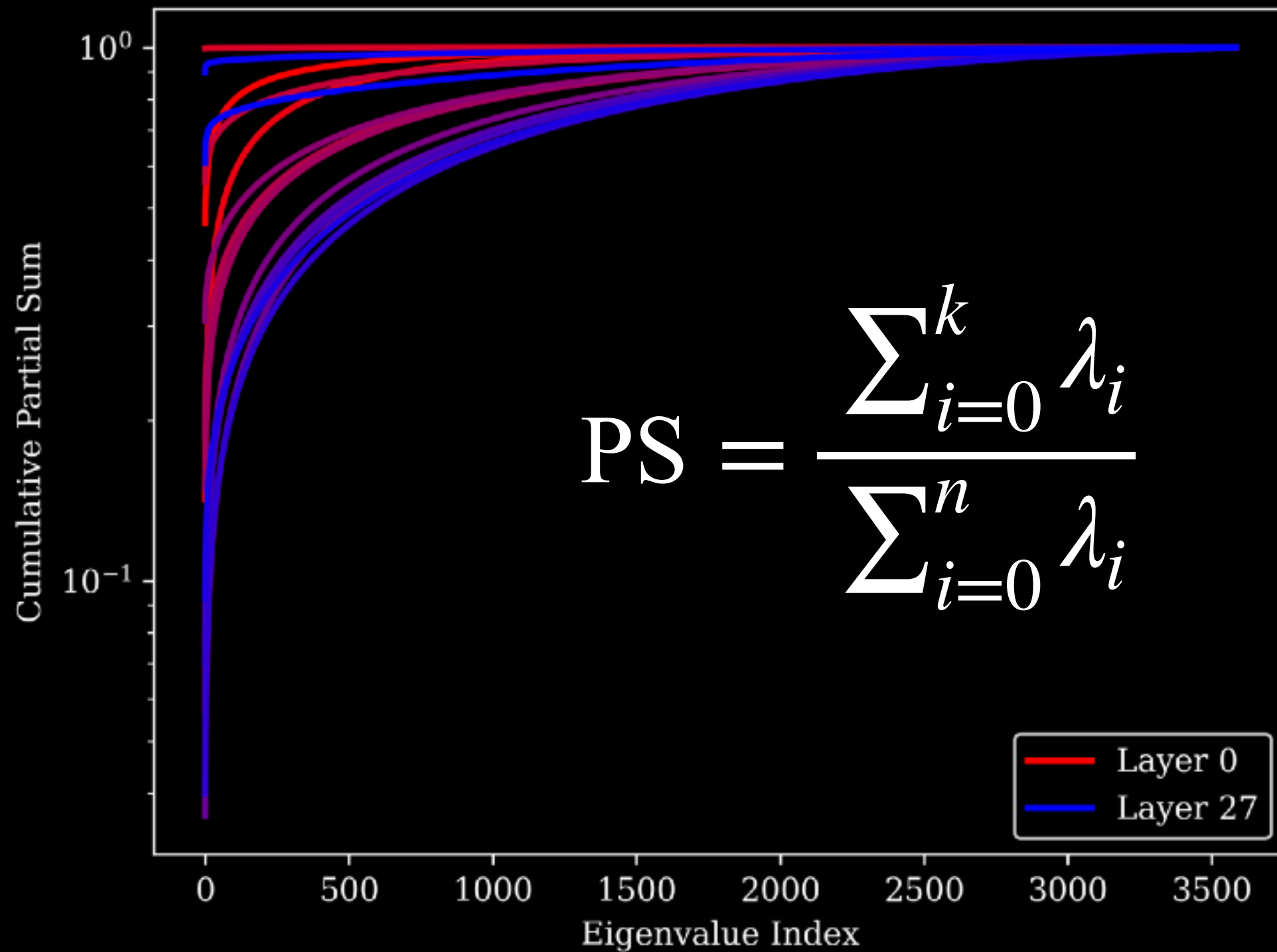
Model Compression

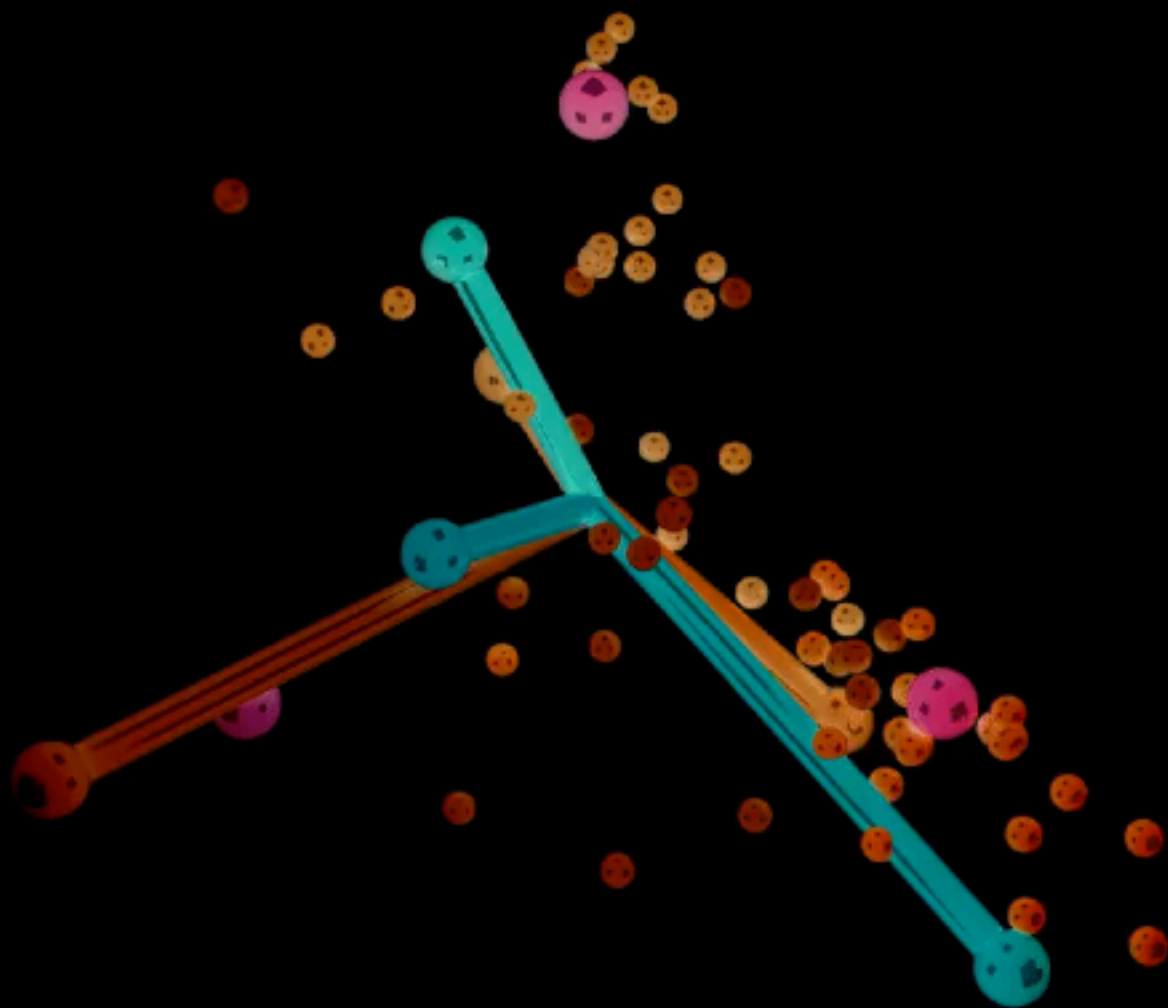
Hyperplane Projection

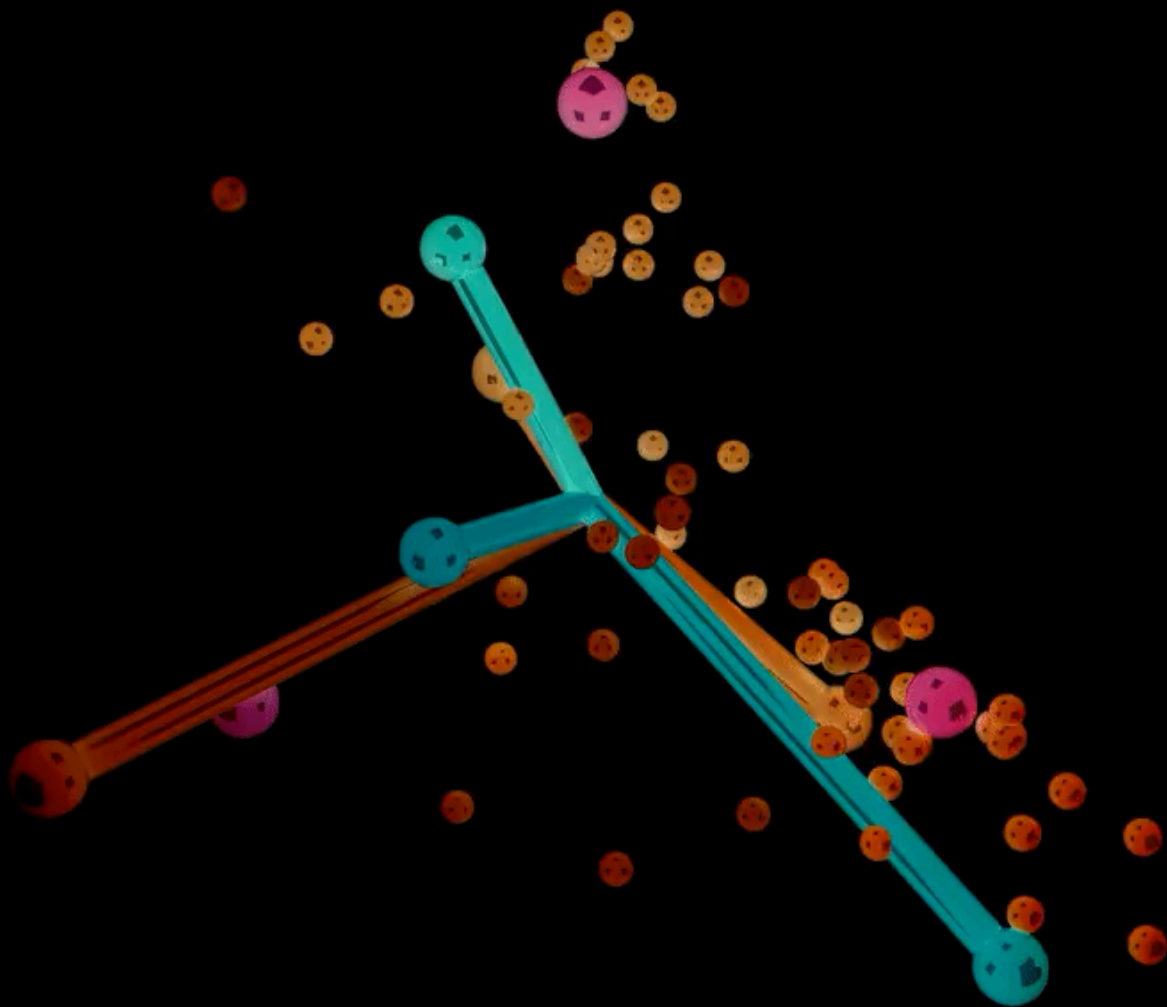
- ↪ Highly targeted model pruning
- ↪ Applications in Edge AI
- ↪ Beyond scope of transformers

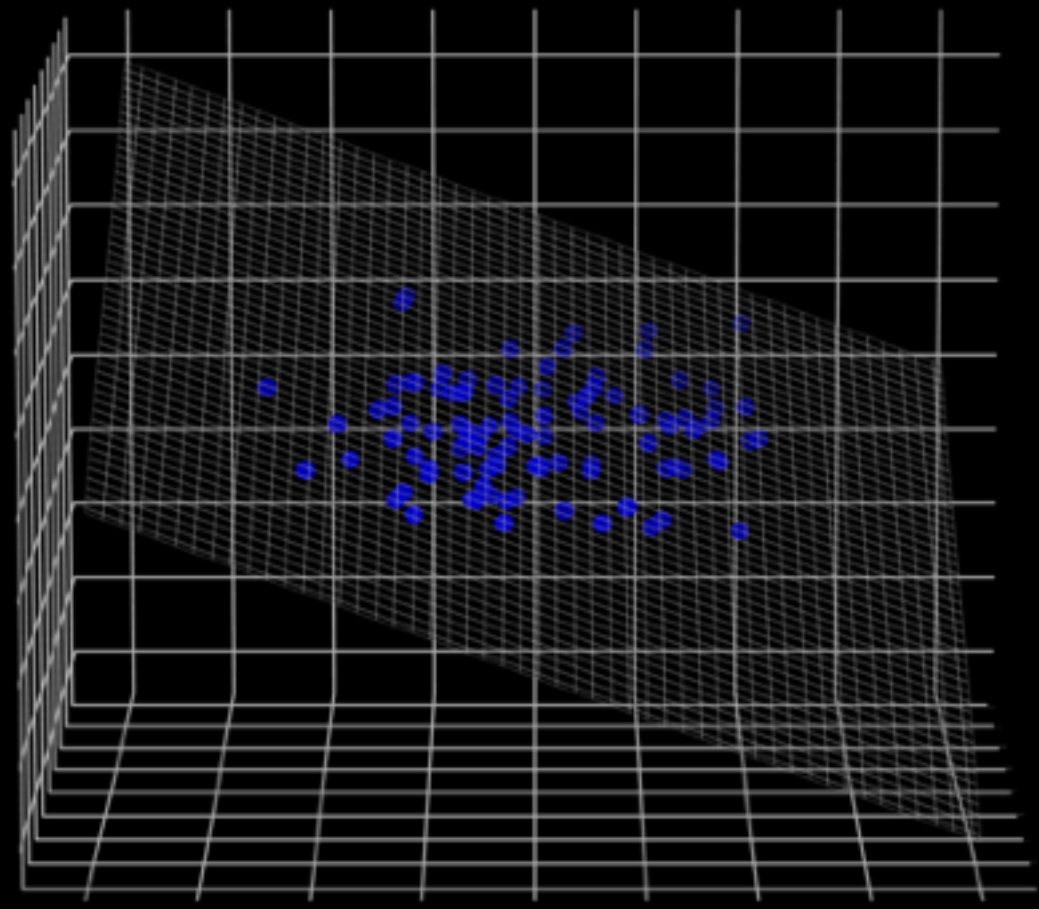
a. Project onto prominent subspace

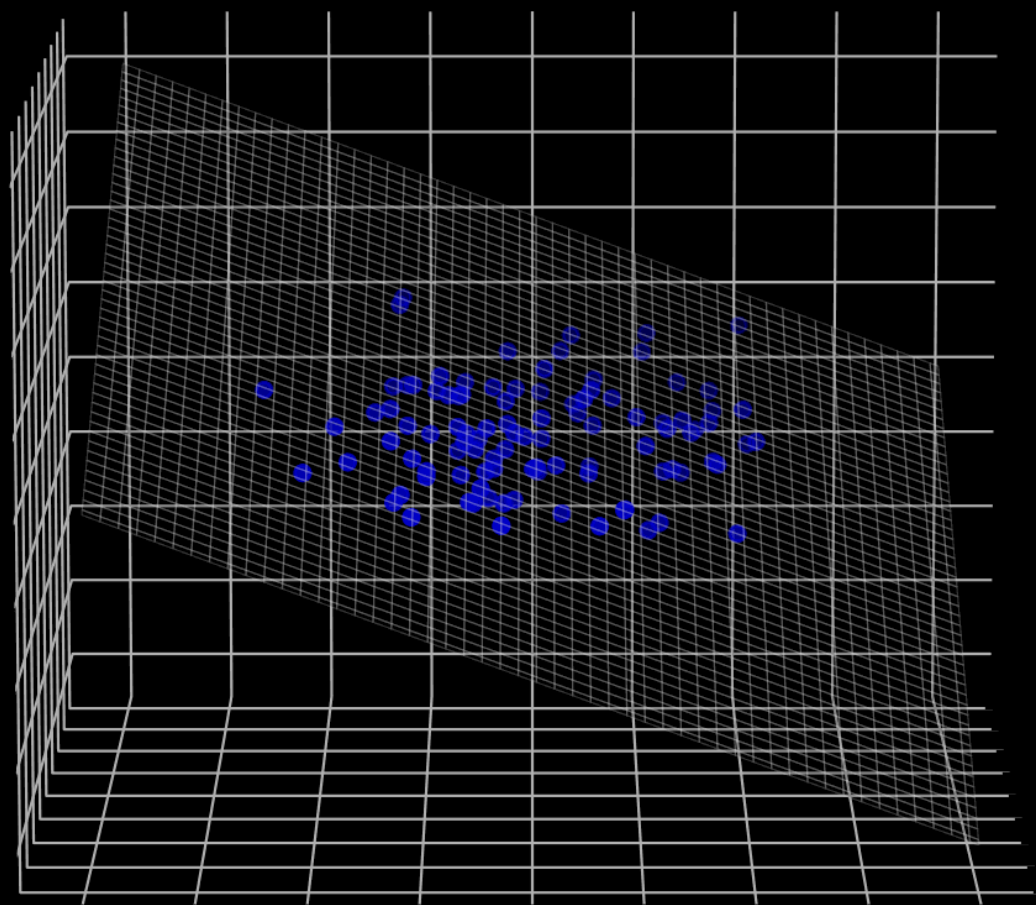
Model: Qwen2.5-Math-7B, Nprompts = 500





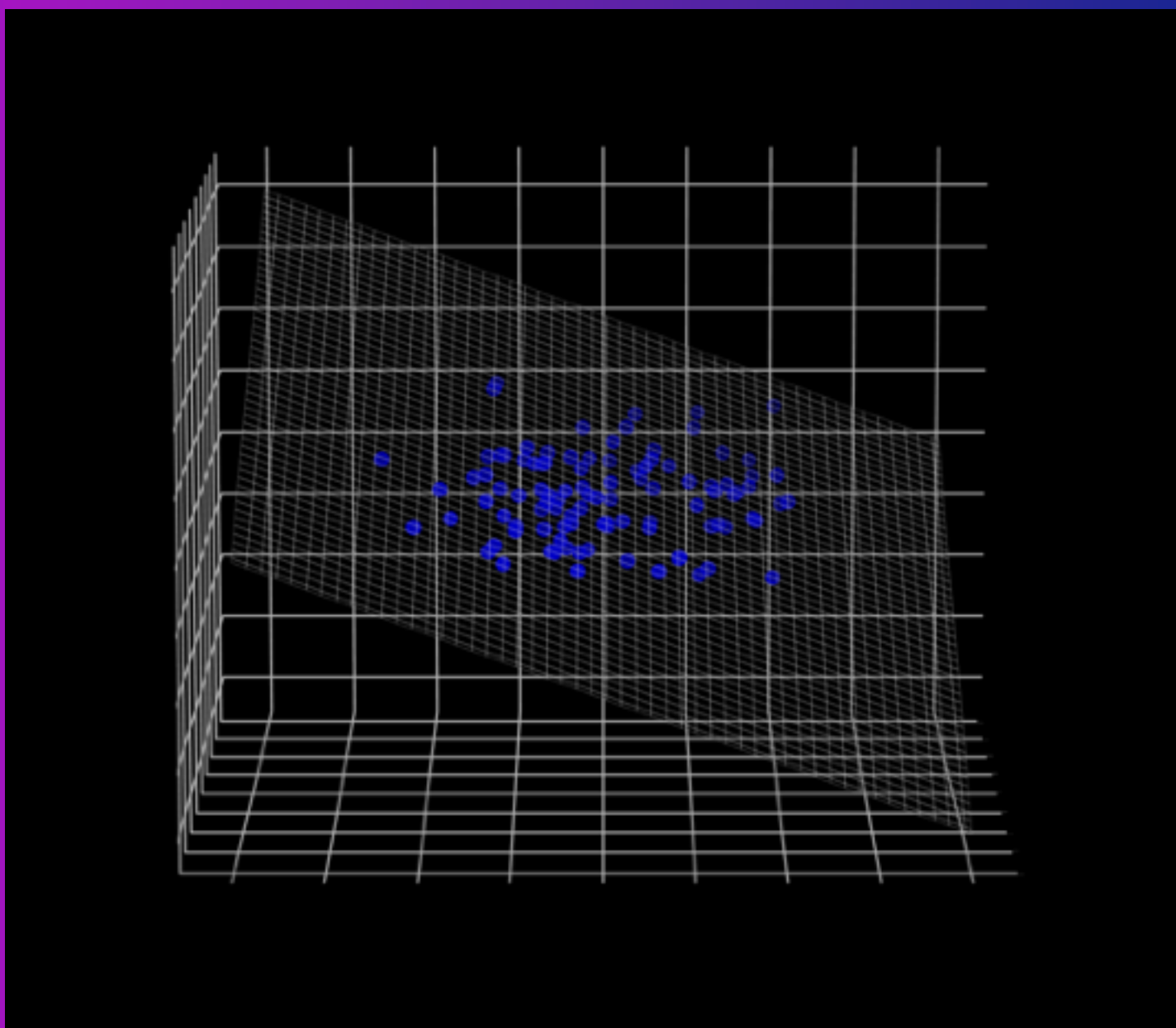






Implications & Applications

Model Compression

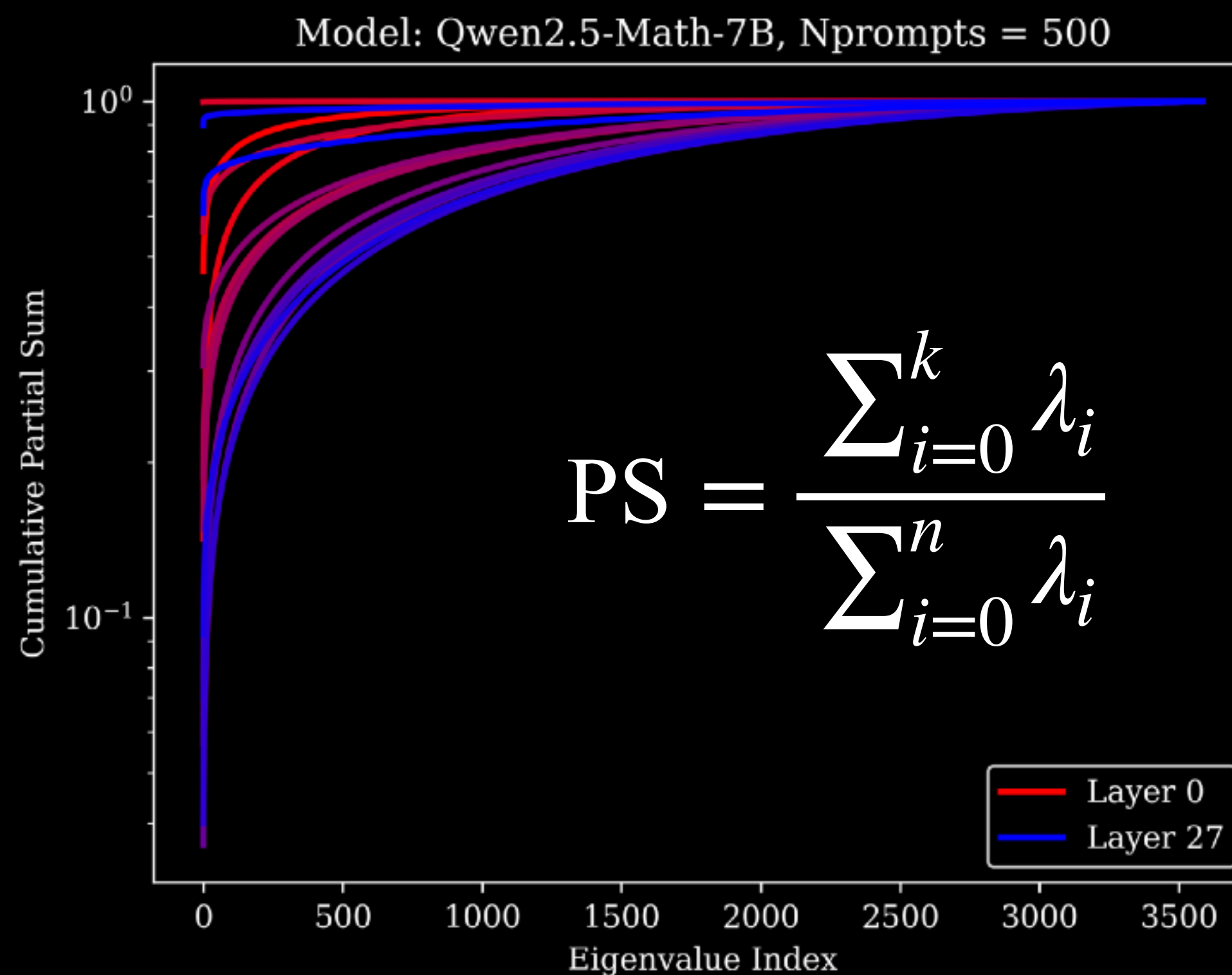


Hyperplane Projection

- ↪ Highly targeted model pruning
- ↪ Applications in Edge AI
- ↪ Beyond scope of transformers

* work in progress *

Idea: Project onto prominent subspace



$$W \leftarrow PWP^T$$

Website



LinkedIn



Thank you!

